Universidad de los Andes
Department of Computer and Systems Engineering

Graduation Project Proposal
Bachelor's in Computing and Systems Engineering

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1 State of the art

1.1 Differential Privacy

Differential privacy describes a promise that is made by a data holder or curator to a data subject: "You will not be affected, adversely or otherwise, by allowing your data to be used in any study or analysis, no matter what other studies, data sets, or information sources, are available".[1] In other words differential privacy is the process that either a data curator or a data holder must do in order to ensure that a single individual's data cannot be extracted from the study as a whole.

Definition 1.1. *(ϵ, δ)-Differential privacy [1]. A randomized algorithm $\mathcal{M}$ will provide (ϵ, δ)-Differential privacy (henceforth (ϵ, δ)-DP) if for any two outputs on two different databases D and D' differing by at most one record, and for all possible outputs A:*

$$Pr[\mathcal{M}(D) \in A] \leq e^{\epsilon} Pr[\mathcal{M}(D') \in A] + \delta$$

In this study we are going to mostly focus around the concept of **Central differential privacy** which states that it is the responsibility of the curator (i.e a central party that receives all the raw data from the participants and adds privacy to them) to assure that the data is going to be differentially private.

1.2 Quantum Information and Computation

Quantum computing significantly advances computational capabilities, performing exponentially faster computation than classical computers.[2] In order to understand some concepts of Quantum differential privacy it is important to define the following notions:

Definition 1.2. *Quantum states: A quantum state of the system X is described by a density matrix ρ whose entries are complex numbers and whose indices (for both its rows and columns) have been place in correspondence with the classical state set Σ*

Definition 1.3. *Mixed state: A mixed state of a quantum system is an ensemble*

$$\mathbb{E} = \{(p_1, |\psi_1\rangle), ..., (p_k, |\psi_k\rangle)\}$$

which in turn can be represented by the density operator

$$\rho_{\mathbb{E}} = \sum_i p_i |\psi_i\rangle \langle\psi_i|$$

thus the resulting $\rho_{\mathbb{E}}$ will be the density matrix representation of the ensemble.

Definition 1.4. *Quantum Channel [3]: A quantum channel describes a linear mapping over density matrices.*

Quantum channels are important as they are the generalization of an operation over a quantum state an in the case of this project we need to use operations such as the depolarizing channel which need this general definition of a channel to be understood.

Definition 1.5. *Trace distance: It is the distance between two density operators and it is defined by:*

$$\tau(\rho, \sigma) = \frac{1}{2} Tr|\rho - \sigma|$$

where $Tr(x)$ is the trace of the density matrix x

Definition 1.6. *Quantum measurement: It is the way to extract information from a quantum system. Matematically, a measurement is modelled as a set of matrices $M = \{K_m\}$ with the condition of*

$$\sum_m K_m^\dagger K_m = I$$

with I being the identity of the density operator (i.e for a qbit $I = 2 \times 2$ identity matrix)

When talking about quantum differential privacy below it won't really matter what the quantum state collapses into. Given this liberty we can define a POVM (Positive Operator-Valued Measure) which is a simplification over the measurement where we get that $M = \{M_m\}$ with $M_m = K_m^\dagger K_m$ $\forall m$ since $p_m = Tr(K_m \rho K_m^\dagger) = Tr(M_m \rho)$

1.3 Quantum Differential Privacy

Given the inherent noise of the current quantum framework, one could guess that it is a perfect scenario for differential privacy. The promise that the classical DP curator makes to the data holder does not change when in the quantum setting however the means to reach that promise do.

In the study of Zhou and Ying [4] they define a framework of differential privacy in the quantum setting. Moreover they examine how different noise mechanisms change the protection of that privacy budget ϵ.

Definition 1.7. *(Quantum differential privacy [4]). Let $d \in (0, 1]$ and $\epsilon, \delta > 0$ be three constants. A quantum operation $\mathcal{E}$ is (ϵ, δ)-differentially private if for every POVM $M = M_m$, for all outputs $S \subseteq Out(M)$ and for all inputs ρ, σ such that $\tau(\rho, \sigma) \leq d$, it holds that:*

$$Pr[\mathcal{E}(\rho) \in_M S] \leq e^\epsilon Pr[\mathcal{E}(\sigma) \in_M S] + \delta$$

In [2] and [5] they further develop on the idea of the noise profile in the quantum setting in order to further the theory onto how noise affects differential privacy in both simulation and real quantum machines. However in my research it has been clear that no explicit framework has been developed to test these hypothesis, as in both works one can only find numerical and analytical results.

2 Objectives

2.1 General Objective

The development of this project will tackle two main objetives: The first will involve an adequate understanding of quantum information and computation as well as the theoretical framework of QDP. The second will be to empirically validate the theoretical (ϵ, δ)-DP guarantees provided by quantum noise mechanisms such as the depolarizing channel, phase-amplitude dampening and GAD, by comparing the privacy budget derived in [4] with the empirical privacy budget measured on a practical quantum simulation on the Pennylane framework.

2.2 Specific Objectives

- Gain an understanding of the classical definition of differential privacy, as well as quantum information and computation to understand quantum differential privacy.

- Create an experiment on the quantum simulation framework Pennylane which will test $(\epsilon, \delta)-DP$ over a real dataset [6]. These results will be explored in further detail in the project's paper.
- Establish whether the theoretical budgets proposed in [4] and [2] agrees or disagrees with the results of the experiments in the simulations for each different noise mechanism.

3 Description of the Experiment

The experiment will follow the following stages:

1. Encode the dataset D with the following rules.
   (a) For categorical variables a set angle will codify each of the three posible scenarios
   (b) For non categorical variables each one will be codified by dense angle encoding thus giving two variables per qbit.
2. Encode the dataset D' which is the same dataset D with one participant removed with the same rules as **1**
3. Simulate a process over the qbits such as aggregation. The result from these processes are going to be the density matrices called ρ and σ
4. Insert noise into ρ and σ
5. Run **Algorithm 2.**[4] on ρ and σ to get the experimental ϵ for that specific run
6. Run the experiment for all D' to get the experimental ϵ for the whole dataset.

4 Expected Results

When doing research or working in a field that involves other people's data it is crucial to set clear bounds as to how that data is going to be collected, managed, distributed, among other considerations. As mentioned earlier in the proposal it is exactly due to this necessity that differential privacy exists in the first place. However, as computers become more and more advanced it is important to explore different possibilities to maintain that promise.

Quantum information and computation is rapidly becoming mainstream as a response to the growing necessity of larger and faster computation. In this sense I believe the investigation will allow for a grounded answer as to if the machines we currently have could realistically manage to protect the privacy of individuals or not.

To achive this the specific results are as follows:

1. Prove that in practice the privacy budget will be preserved within reasonable proximity to the theoretical budget in an experimental setup for the different noise injections sources and classical data sources.
2. Provide the source code for the experimental setup in Pennylane with python.
3. Write a scientific article in IEEE format. The information within this article should demonstrate my understanding of the topic as it will contain the whole process of investigation, creation and result analysis of this project.
4. Design a poster which will summarize the most important parts of the project.

5 Ethical Considerations

This investigation will use an adaptation [7] of the Statlog German Credit Data dataset [6], originally compiled by Prof. Hans Hofmann in 1994 which has a public CC BY 4.0 license thus allowing the use of the dataset for any purpose as long as the appropriate credit is provided.

This dataset was selected as it encompasses the exact neccessity that differential privacy is trying to protect: Given sensitive data from individuals (sex/marital status, foreign worker status, etc.), what is the privacy budget that can be offered to the individuals in exchange of the use of their information in the dataset.

6 Activity Calendar

Week #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Activity																
Investigation on the viability of the project and scope																
Write the proposal of the project																
Search and data analysis of the experimental data to be used in the experiments																
Write the introductory section of the paper (Introduction, Theoretical Framework)																
Real data experiment development in Pennylane																
Analysis of the first two experiment's results																
Write the first revision of the paper																
Adjustments to the experiments with respect to the result of the analysis																
Write the second revision of the paper																
Write and design the other deliverables (Poster, Project document, etc)																

References

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[6] H. Hofmann, “Statlog (German Credit Data),” *UCI Machine Learning Repository*, 1994. [Online]. Available: https://doi.org/10.24432/C5NC77.

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